CLASS 8 — SCIENCE · NCERT-TYPE UNIT TEST SET 3

Chapters 5–9: **Exploring Forces, Pressure, Winds, Storms and Cyclones, Particulate Nature of Matter, Nature of Matter: Elements, Compounds and Mixtures, The Amazing World of Solutes, Solvents and Solutions** | RBSE Pattern | Session 2026–27

Time: 2 Hours Name: _____ Roll No.: _____ **Maximum Marks: 50**

General Instructions:

1. All questions are compulsory.
2. Q. 1–15 are Objective Type (1 mark each).
3. Q. 16–20 are Very Short Answer Type (2 marks each).
4. Q. 21–25 are Short Answer Type (3 marks each).
5. Q. 26–27 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 24–25 and Q. 26–27.
7. Write the question number before each answer.
8. NCERT-type: questions in the style of the textbook examples and exercises.

SECTION — A (Objective Type Questions) 15 × 1 = 15

Choose the correct option and write it in your answer book.

1. A force can change: 1

(a) the speed, direction or shape of an object
(b) only the mass
(c) only the colour
(d) only the temperature

2. A force of 100 N acts on an area of 2 square metres. The pressure is: 1

(a) 98 Pa **(b)** 0.02 Pa **(c)** 200 Pa **(d)** 50 Pa

3. The particles of matter are: 1

(a) stuck together permanently **(b)** always at rest in solids
(c) constantly moving **(d)** completely still

4. Which of the following is a compound? 1

(a) CO_2 **(b)** Fe **(c)** O_3 **(d)** N_2

5. In a solution of sugar in water, water is the: 1

(a) solvent **(b)** solute **(c)** alloy **(d)** suspension

6. Compared with sliding friction, rolling friction is: 1

(a) twice as much **(b)** less **(c)** the same **(d)** more

7. Winds are caused by: 1

(a) rotation of fans
(b) moon only
(c) rain
(d) differences in air pressure due to uneven heating

8. The temperature at which a solid melts at normal pressure is its: 1

(a) freezing point of gases **(b)** room temperature
(c) melting point **(d)** boiling point

9. Metals produce a ringing sound when struck. This property is called: 1

(a) ductility **(b)** sonority **(c)** malleability **(d)** lustre

10. A mixture in which particles settle down on standing is a: 1

(a) suspension **(b)** compound **(c)** colloid **(d)** solution

11. Unlike charges: 1

(a) do not interact **(b)** vanish

(c) attract each other **(d)** repel each other

12. A tropical cyclone forms over: 1

(a) mountains **(b)** poles **(c)** cold deserts **(d)** warm oceans

13. Evaporation causes: 1

(a) cooling **(b)** heating **(c)** burning **(d)** no change

14. Air is a: 1

(a) single element **(b)** single compound **(c)** pure substance **(d)** mixture of gases

15. 10 g of sugar is dissolved in 90 g of water. The mass of solution is: 1

(a) 80 g **(b)** 10 g **(c)** 100 g **(d)** 90 g

SECTION — B (Very Short Answer Type) 5 × 2 = 10

Answer in about 20–30 words each.

16. State the law of charges. 2

17. What is the 'eye' of a cyclone? 2

18. Name the process in the formation of dew. 2

19. Why is graphite used in pencils and as electrodes? 2

20. Find the mass percentage of a solution with 5 g salt in 45 g water. 2

SECTION — C (Short Answer Type) 5 × 3 = 15

Answer in about 40–60 words each.

21. State three uses of solutions in daily life. 3

22. Describe the types of friction with an example for each. 3

23. Explain why it is easier to cut with a sharp knife than a blunt one. 3

24. Describe an activity to show diffusion in liquids. 3

OR

List three physical properties of non-metals. 3

25. Explain how you prepare a saturated solution of common salt. 3

OR

Explain how a body can get charged by rubbing. Give an example. 3

SECTION — D (Long Answer Type) 2 × 5 = 10

Answer in about 100–150 words each.

26. Explain atoms and molecules with the help of H_2, O_2, H_2O and CO_2, giving the number of atoms in each. 5

OR

Describe the types of solutions on the basis of the physical states of solute and solvent with an example each. 5

27. Explain the types of forces: contact and non-contact, with at least two examples for each. **5**

OR

Explain how pressure in liquids depends on depth and how it acts in all directions. Describe an activity. **5**

END OF QUESTION PAPER **Total: 50 Marks**

MASTERMIND